

PROJECT PRESENTATION: MINI EMPLOYEE MANAGEMENT SYSTEM (EMS)

Slide 1: Title & Welcome

 Modern Full-Stack Employee Management System

Final Project Presentation - Full Stack Development Training Program

- **Presenter:** Internship Student
 - **Academic/Internship Partner:** i-SOFTZONE
 - **Focus:** Cloud Deployment, Schema Optimization, and Business Logic Automation
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Slide 2: Problem Statement & Context

 The Challenge of Scalable HR Operations

Traditional employee registry trackers face major scaling pain points:

- **Manual Data Silos:** Inefficient tracking of departments, skillset matrices, and documents.
 - **Concurrency Issues:** Race conditions during leave balance calculations and approvals.
 - **Performance Degradation:** Slow queries and table scans as employee profiles and attendance histories grow.
 - **Security Gaps:** Lack of rigid Role-Based Access Controls (RBAC).
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Slide 3: The Solution

 Mini EMS Platform Overview

An integrated, low-latency web platform offering automated workflows:

- **Unified Directory:** Interactive employee profiles complete with skill keywords and attachments.
 - **Atomic Leave Engine:** Interactive database transactions handling leave approvals and balance deductions.
 - **Inventory Control:** Complete hardware asset issuance, history logs, and return operations.
 - **Audit Trail & Alerts:** Real-time user notifications and automated change-history audit logging.
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Slide 4: Technology Stack

 Industry-Standard Technologies

A decoupled, performant tech stack optimized for seamless cloud deployments:

- **Frontend:** React.js, Vite, Tailwind CSS, Lucide Icons, Recharts (for dashboard analytics).
- **Backend:** Node.js, Express, Winston Logger, Joi (validation schemas), Swagger UI.

- **ORM:** Prisma Client (type-safe queries, migration control).
 - **Database:** Cloud PostgreSQL (Neon DB).
 - **Hosting:** Vercel (Frontend SPA) & Render (Backend API Web Service).
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Slide 5: System Architecture

🌐 Core Data & Deployment Flow

```
[ React SPA Client (Vercel) ]
      |
      v (Secure HTTP & JWT Bearer token headers)
[ Express API Server (Render) ]
      |
      v (Prisma Client Type-Safe Queries)
[ PostgreSQL Database (Neon Cloud) ]
```

- **Fetch Interceptors:** Automatically inject JWT access tokens and dynamically rewrite URL routes for production API environments.
 - **SPA Routing Fix:** Configured `vercel.json` rewrite rules to prevent 404 router errors on page refresh.
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Slide 6: Database Optimization & Indexing

⚡ Engineered for Performance

To support rapid search and joins, custom index patterns were implemented on foreign keys:

- **Employee(department_id):** Optimizes department roster loads and joins.
 - **EmployeeDocument(employee_id):** Speeds up retrieval of profile attachments.
 - **EmployeeSkill(employee_id, skill_id):** Resolves many-to-many lookups.
 - **LeaveApplication(employee_id, status):** Optimizes dashboard filters.
 - **Notification(user_id, is_read):** Powers real-time alert counters.
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Slide 7: Transaction Safety & Concurrency

🔒 Atomic Transactions (ACID Rules)

For leave requests final review, the system avoids "double-deduction" or race conditions using Prisma's interactive transaction API:

```
await prisma.$transaction(async (tx) => {
  // 1. Approve Leave application status
  // 2. Decrement available days from Employee's Balance
  // 3. Log event into ApprovalHistory audit table
});
```

- **All-or-Nothing Execution:** If any step fails, the database automatically rolls back, maintaining absolute data consistency.
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Slide 8: Live Features Demonstration

System Capabilities

- **Dynamic Analytics Dashboard:** Visualizes department counts and leave ratios using Recharts.
 - **Fuzzy Directory Search:** Pagination and name sorting on employee directory.
 - **Shift Tracking (Clock In/Out):** Automatically logs attendance and calculates daily hours.
 - **Master Records Config:** Add departments and skills dynamically to update the company registry dropdowns.
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Slide 9: Deployment Coordinates

Live Submission URLs

- **GitHub Repository:** <https://github.com/Atharva-28-04/Employee-Management-System.git>
 - **Live Frontend Website:** <https://mini-ems-frontend.vercel.app> (Vercel URL)
 - **Live Backend API Services:** <https://mini-ems-backend.onrender.com/api/v1/health> (Render URL)
 - **Cloud Database Engine:** Neon PostgreSQL.
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Slide 10: Conclusion & Thank You!

Training Completed

This project illustrates a complete, enterprise-grade deployment workflow mirroring standard software companies' software delivery cycles.

- **Key Takeaway:** Achieved clean separation of concerns, secure access controls, optimized databases, and portable deployments.
- **We wish to express our appreciation to the i-SOFTZONE team for their consistency, guidance, and training support!**

Questions & Feedback are Welcome! 